BEE 4750 Homework 2: Systems Modeling and Simulation

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Due Date

Thursday, 09/25/25, 9:00pm

Overview

Instructions

- Problem 1 asks you to draw a systems diagram and identify the type of a feedback.
- Problem 2 asks you to model contaminant concentrations in a river and use simulation to compare the concentrations to a regulatory standard.
- Problem 3 asks you to explore the implications of an ice-albedo feedback in the Earth's climate system by understanding the equilibria of the modeled system and their stabilities.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```
  Activating project at `~/hw2-clairej`
Precompiling project...
 Warning: attempting to remove probably stale pidfile
   path = "/Users/clairejacobson/.julia/compiled/v1.11/SortingAlgorithms/6dCmw_qNIpJ.ji.pidf
 @ FileWatching.Pidfile ~/.julia/juliaup/julia-1.11.5+0.x64.apple.darwin14/share/julia/stdli
 Warning: attempting to remove probably stale pidfile
   path = "/Users/clairejacobson/.julia/compiled/v1.11/Accessors/XelUh_qNIpJ.ji.pidfile"
 @ FileWatching.Pidfile ~/.julia/juliaup/julia-1.11.5+0.x64.apple.darwin14/share/julia/stdli
   3351.5 ms  Graphite2_jll
   4132.4 ms  Libmount_jll
   5753.9 ms  OpenSSL_jll
   9049.5 ms  SortingAlgorithms
   5130.3 ms  EpollShim_jll
   6124.3 ms  LLVMOpenMP_jll
   5514.3 ms  Bzip2_jll
   5521.7 ms  Xorg_libICE_jll
   5674.0 ms  Xorg_libXau_jll
   5895.4 ms  libpng_jll
  16493.5 ms  Accessors
   6154.2 ms  libfdk_aac_jll
   6355.5 ms  LAME_jll
   5067.1 ms  LERC_jll
   3914.2 ms  fzf_jll
   4824.3 ms  JpegTurbo_jll
   5061.7 ms  XZ_jll
   5260.4 ms  Ogg_jll
   4844.8 ms  mtdev_jll
   5022.8 ms  Xorg_libXdmcp_jll
   4822.8 ms  x265_jll
   4799.6 ms  x264_jll
   4396.1 ms  Zstd_jll
   5201.0 ms  libaom_jll
   5524.1 ms  Expat_jll
   5295.0 ms  LZO_jll
   5149.5 ms  Opus_jll
   4902.3 ms  Xorg_xtrans_jll
   4717.3 ms  libevdev_jll
   4688.1 ms  Libiconv_jll
 Warning: attempting to remove probably stale pidfile
   path = "/Users/clairejacobson/.julia/compiled/v1.11/RecipesBase/8e2Mm_qNIpJ.ji.pidfile"
 @ FileWatching.Pidfile ~/.julia/juliaup/julia-1.11.5+0.x64.apple.darwin14/share/julia/stdli
   4952.9 ms  Libffi_jll
   4760.4 ms  eudev_jll
```

```
 5078.3 ms   Libuuid_jll
 5426.8 ms   FriBidi_jll
Warning: attempting to remove probably stale pidfile
  path = "/Users/clairejacobson/.julia/compiled/v1.11/FilePathsBaseMmapExt/bw4lJ_qNIpJ.ji.p
@ FileWatching.Pidfile ~/.julia/juliaup/julia-1.11.5+0.x64.apple.darwin14/share/julia/stdli
 5674.6 ms   InlineStrings → ParsersExt
Warning: attempting to remove probably stale pidfile
  path = "/Users/clairejacobson/.julia/compiled/v1.11/FilePathsBaseTestExt/5f6TB_qNIpJ.ji.p
@ FileWatching.Pidfile ~/.julia/juliaup/julia-1.11.5+0.x64.apple.darwin14/share/julia/stdli
 6117.4 ms   FilePathsBase → FilePathsBaseMmapExt
10394.3 ms   RecipesBase
13251.1 ms   StringManipulation
 7310.5 ms   FilePathsBase → FilePathsBaseTestExt
 6182.3 ms   Pixman_jll
 6140.0 ms   FreeType2_jll
21164.3 ms   ColorSchemes
12757.3 ms   OpenSSL
14233.8 ms   StatsBase
 4263.9 ms   Xorg_libSM_jll
 5343.9 ms   Accessors → LinearAlgebraExt
 4551.8 ms   Accessors → TestExt
 4717.7 ms   Accessors → UnitfulExt
 3571.7 ms   Xorg_libxcb_jll
 8341.4 ms   JLFzf
 6455.7 ms   libvorbis_jll
 9942.8 ms   Ghostscript_jll
 4210.3 ms   Dbus_jll
 3617.4 ms   Wayland_jll
 4167.6 ms   libinput_jll
 6809.7 ms   GettextRuntime_jll
12461.5 ms   Libtiff_jll
 6582.6 ms   WeakRefStrings
11834.2 ms   Fontconfig_jll
 3058.6 ms   Xorg_xcb_util_jll
 3325.1 ms   Xorg_libX11_jll
15683.9 ms   Roots
 8932.7 ms   Glib_jll
17163.3 ms   Latexify
 3611.2 ms   Xorg_xcb_util_image_jll
 3482.7 ms   Xorg_xcb_util_keysyms_jll
 3487.1 ms   Xorg_xcb_util_renderutil_jll
 3631.9 ms   Xorg_xcb_util_wm_jll
50678.2 ms   PlotUtils
```

```
  3480.0 ms    Xorg_libXrender_jll
  2940.7 ms    Xorg_libXext_jll
  2799.7 ms    Xorg_libXfixes_jll
  3626.6 ms    Xorg_libxkbfile_jll
  6585.5 ms    Roots → RootsUnitfulExt
  9013.0 ms    UnitfulLatexify
  6840.8 ms    Latexify → SparseArraysExt
  3572.3 ms    Xorg_xcb_util_cursor_jll
 18588.5 ms    PlotThemes
  4560.6 ms    Xorg_libXinerama_jll
  5024.8 ms    Xorg_libXrandr_jll
 30728.2 ms    RecipesPipeline
  4705.8 ms    Libglvnd_jll
101132.7 ms    HTTP
  4161.4 ms    Xorg_libXcursor_jll
  5453.1 ms    Xorg_libXi_jll
  4204.2 ms    Xorg_xkbcomp_jll
 12959.6 ms    Cairo_jll
  4586.0 ms    Xorg_xkeyboard_config_jll
 90973.7 ms    CSV
  3589.3 ms    xkbcommon_jll
  6007.8 ms    HarfBuzz_jll
  3620.8 ms    Vulkan_Loader_jll
  6943.4 ms    libass_jll
  7183.9 ms    Pango_jll
  3591.3 ms    libdecor_jll
  3481.0 ms    GLFW_jll
144264.6 ms    PrettyTables
 16410.2 ms    FFMPEG_jll
 23788.0 ms    Qt6Base_jll
  5596.8 ms    FFMPEG
  4258.3 ms    Qt6ShaderTools_jll
  6044.8 ms    GR_jll
  6505.7 ms    Qt6Declarative_jll
  1706.5 ms    Qt6Wayland_jll
 10227.0 ms    GR
112494.8 ms    DataFrames
  5259.5 ms    Latexify → DataFramesExt
152339.3 ms    Plots
  8117.0 ms    Plots → IJuliaExt
  9143.0 ms    Plots → UnitfulExt
 110 dependencies successfully precompiled in 410 seconds. 116 already precompiled.
```

```julia
using Plots
using LaTeXStrings
using CSV
using DataFrames
using Roots
```

Problems (Total: 30 Points)

Problem 1 (5 points)

Draw a systems diagram for the relationship between global mean temperature, atmospheric CO2 concentrations, and ocean CO2 concentrations. What are the signs of the interactions between these components and why? What does this suggest about the overall feedback between temperature and the ocean carbon cycle?

> **Tip**
>
> Think about Henry's law for CO2.

Problem 2 (15 points)

A river which flows at 10 km/d is receiving discharges of wastewater contaminated with CRUD from two sources which are 15 km apart, as shown in the Figure below. CRUD decays exponentially in the river at a rate of 0.36 d^{-1} and is deposited by the atmosphere along the river at a rate of 54 kg/km/d.

Schematic of the river system in Problem 2

Problem 2.1

Draw a systems diagram with the relevant control volume(s) denoted and any relevant in/outflows between these boxes. How did you decide how many boxes were needed?

Problem 2.2

Develop a model for the concentration of CRUD downriver by formulating and solving the appropriate differential equation(s) analytically.

> **Tip**
>
> Formulate your model in terms of distance downriver, rather than leaving it in terms of time from discharge.

Problem 2.3

Determine if the system in compliance with a regulatory limit of 2.3 kg/(1000m^3). You can do this analytically or computationally.

Problem 3 (10 points)

In class, we discussed the ice-albedo feedback and its possible influence on melting the hypothesized "Snowball Earth". In this problem, we'll introduce a simple model of the Earth's energy balance with includes this feedback and examine the stability of the climate.

This simple model of the energy balance (averaged over the entire planet) is:

$$\underbrace{C\frac{dT}{dt}}_{\text{change in heat}} = \underbrace{\frac{(1-\alpha)S}{4}}_{\text{incoming radiation}} - \underbrace{(A-BT)}_{\text{outgoing radiation}} + \underbrace{a\ln\left(\frac{[CO_2]}{[CO_2]_{PI}}\right)}_{\text{greenhouse effect}}, \tag{1}$$

where:

- T is the Earth's global mean temperature (in $^\circ$C);
- C is the heat capacity of the atmosphere and shallow ocean, taken to be 51 J/m^2/$^\circ$C;
- α is the planetary albedo, or the fraction of incoming radiation reflected by the Earth, which has a present-day value of approximately 0.3;
- S is the solar constant, or the amount of solar radiation received by the Earth averaged over area, which has a present-day value of 1368 W/m^2 but during the Neoproterozoic Era had a value of 1272 W/m^2 (this is divided by four because the Earth is a sphere but S is the radiation captured by a disc with radius equal to that of the Earth);
- A and B are coefficients from the linearization of outgoing radiation physics, and have corresponding values $B = -1.3$ W/m^2/$^\circ$C (estimated from a number of lines of evidence about the sensitivity of outgoing radiation to temperature; this is negative due to a sign convention about the direction of incoming vs. outgoing radiation) and $A = 221.2$ W/m^2 (estimated by assuming the pre-industrial temperature of 14°C was stable without anthropogenic greenhouse gas emissions).

We will ignore the greenhouse effect term as we are considering the Earth system well before humans were around.

Problem 3.1

Discretize the climate model (Equation 1) using forward Euler integration and a time step of $\Delta t = 1$ yr.

Problem 3.2

Rather than assuming a constant value for the albedo α, we will represent the ice-albedo feedback by letting α depend on T:

$$\alpha(T) = \begin{cases} \alpha_i & \text{if } T \leq -10^\circ\text{C} \\ \alpha_i + (\alpha_0 - \alpha_i)((T+10)/20) & \text{if } -10^\circ\text{C} \leq T \leq 10^\circ\text{C} \\ \alpha_0 & \text{if } T \geq 10^\circ\text{C}. \end{cases}$$

Let $\alpha_i = 0.5$ and $\alpha_0 = 0.3$. Simulate the simple climate model using the Neoprotereozoic value of S with temperature-varying albedo for initial values of T spanning $-60^\circ\text{C} \leq T_0 \leq 30^\circ\text{C}$ over a period of 200 years. Plot the temperature trajectories. How many equilibria are there and what are their stabilities?

Problem 3.3

One might hypothesize that one cause for the transition from Snowball Earth (the stable frozen equilibrium from Problem 3.2) was an increasing amount of incoming solar radiation (as expressed by an increase in S). Examine this hypothesis using our simple model. Is this factor enough to cause the planet to warm to the pre-industrial temperature of 14°C?

References

List any external references consulted, including classmates.